



A density map regression method and its application in the coal flotation froth image analysis

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ABSTRACT

The surface feature of flotation froth is an indicator of the flotation process state. Image-based methods have long been considered as an indirect detector to access flotation working conditions. However, large and small bubbles stick together, resulting in shadows, occlusions and defocus problems in the flotation froth images acquired in the field. These problems lead to resistance to the accurate extraction of morphological features. This paper attempts to analyze the morphology of froth images from the perspective of bubble distribution. The proposed framework generates density maps measuring the sparsity of bubbles, which also serves as a meter to imply the morphology of froth images. In order to improve the quality of the regressed density map, label normalization was proposed to calibrate ground truth during the dataset production phase; the deconvolution module was introduced to the network to gain smoother bubbles boundaries; the loss selective drop mechanism was used to mitigate the negative impact of annotation deviation during the model training. The effectiveness of each module in the framework was verified by a series of ablation experiments on the coal flotation froth dataset. The experimental data shows that the error in measuring bubble mean size is 1.7%, the error in measuring ratio of the area between large and small bubbles is 0.8%, and the accuracy of measuring the spatial distribution of large and small bubbles is satisfactory.

1. Introduction

Flotation is a critical technology in fine coal separation, used to separate coal and gangue. The current monitoring and control of flotation mainly rely on the operators observing the froth in the flotation tank and intuitively regulating its flotation conditions. This heavily subjective process is inconducive to improving flotation efficiency [1]. Intelligent regulation of the flotation process was recognized to enhance fine coal yield and recovery, but this vision has long been hampered due to the lack of appropriate sensors. Previous research has established that the surface appearance of flotation froth is closely related to the flotation process working conditions and directly reflects the rationality of flotation process regulation [2–4]. Therefore, accurate acquisition of surface features of flotation froth is a prerequisite for intelligent regulation of the flotation process.

The features of flotation froth are often described by morphological and color characteristics. Compared to metal ore flotation froth, the color of coal flotation froth does not change significantly, and its color characteristics are difficult to reflect the working conditions significantly [5]. Therefore, morphological information such as the size and distribution of coal flotation froth becomes more imperative. Specifically, there is a strong correlation between the mean size of bubbles and process conditions, such as the addition of flotation reagents [6]. Nevertheless, this correlation may sometimes be inapplicable, as shown in Fig. 1(a). Although two images have the same mean size (i.e., 1424 pixels), they work at different reagent levels. We argue that more detailed metrics are needed to complement the mean size to measure the morphological characteristics of the bubbles. For instance, the ratio of the area between large and small bubbles is an indicator that can assist in identifying the health status of the reagent addition. The spatial

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distribution of large and small bubbles can identify the accumulation of small bubbles caused by the problem of reagent stirring. Combining the mean size with the two bubble indicators mentioned above can characterize the state of coal flotation froth in a detailed manner.

Obtaining accurate surface features of flotation froth requires practical feature extraction algorithms. Although the study about flotation bubble images lasted a long time, very few successful implementations that work reliably over long periods were found [7–9]. Steady progress has been made in analyzing froth image features due to the emergence of many froth image studies based on advanced classification and segmentation algorithms [10–12]. However, the flotation froth is in a non-uniform and non-directional motion at all times, and bubbles stick to each other, resulting in shadow obscuration and motion blurring problems (as shown in Fig. 1(b)). These problems lead to resistance to accurate extraction of morphological features, and current methods show their limitations. In particular, the segmentation-based algorithm is difficult to avoid the under-segmentation problem due to blurring/shadowing even under the constraint of extremely detailed manual annotation (Fig. 1(c)). Therefore, the bottleneck that currently restricts the application of research results in practical production is the failure to extract accurate and robust flotation froth features even after investing a large amount of manual annotation cost.

In response to the above problems, we attempt to analyze the morphological characteristics of froth images from the perspective of spatial bubble distribution. We used density maps to characterize the sparsity of the spatial bubble distribution. This paper presents a framework for extracting morphological characteristics of flotation froth based on density map regression. Specifically, the idea of density map regression was introduced from the crowd counting task [13], the bubbles were annotated in points form, and the density map was generated according to the relative position between bubbles. The density map was used as ground truth to feed into the network for training along with the bubble image. Regarding the regression network, we proposed label normalization to calibrate the ground truth during the dataset production phase, which improves the quality of the regressed density map. The deconvolution module was embedded into the network to gain smoother bubble boundaries. The loss selective drop

mechanism was used to mitigate the negative impact of annotation deviation during the model training. The effectiveness of each module in the framework has been proved through a series of ablation experiments on the industrial coal flotation froth dataset.

The paper is organized as follows: the related work about flotation froth image analysis and density map generation are reported in Section 2; the details of the density map regression framework are explained in Section 3; the acquisition of the flotation dataset and the implementation of the model are described in subsection 4; experimental results and discussion are presented in subsection 5; Section 6 summarized the overall work.

2. Related works

2.1. Feature extraction from froth images

Whether in the field of metal ore flotation or coal flotation, it is accepted that the surface characteristics of flotation froth directly reflect the state of flotation [14]. Therefore, many researchers have worked on directly constructing end-to-end models of flotation froth images and flotation conditions. In this process, early researchers classified the flotation froth images under different working conditions by describing the global color features [15] and texture information [16,17] of the images within the macroscopic field of view [18,19]. With the development of deep learning, CNN and RNN have demonstrated powerful feature representation in the spatial domain [20] and temporal domain [21], respectively. Some researchers used classical image classification networks (i.e., AlexNet, VGG, ResNet, etc.) for flotation froth image classification, showing positive effects [22–24]. Ai et al. [25] developed a two-stream network to extract the froth surface and movement features, and the attention mechanism was employed to improve its performance. Wang et al. [26] first used GAN to extract global froth image features without supervision, and this method improved the semantic interpretability of features. However, the global features-based method for classifying flotation conditions is too rough to describe the independent bubbles in more detail completely.

The segmentation algorithm is the mainstream method for studying

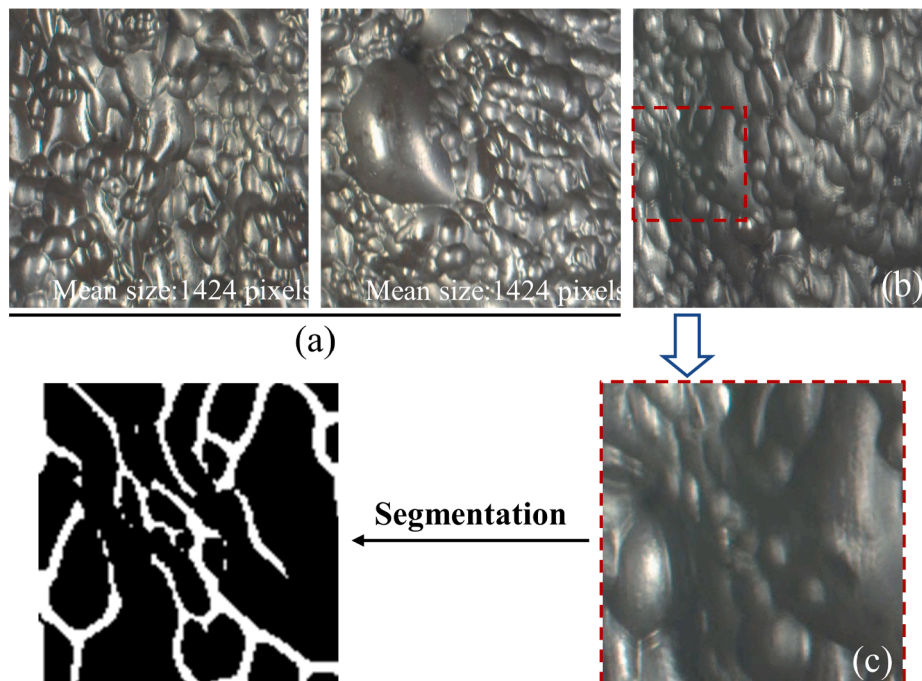


Fig. 1. Example of extracting morphological characteristics of flotation froth images. (a) The mean size of both flotation froth images is 1424 pixels, but a vast bubble is present in the right image.

the surface features of each bubble. Among them, most morphological studies are based on the watershed algorithm, combining flotation froth characteristics. Improvements mainly focus on (a) the boundary constraints [27], (b) seed point selection [28], and (c) gradient calculation [29]. As semantic segmentation algorithms based on deep learning demonstrated strong foreground and background separation capabilities in numerous fields [30,31], the researcher has also introduced them into the flotation froth image analysis field. Zhang and Xu [32] employed U-Net to segment flotation froth images. Liu et al. [33] designed a semantic segmentation network based on the idea of a fully convolutional network and made improvements in deblurring and feature fusion. However, due to the unavoidable shadows and blurring problems during image acquisition, neither morphology-based watershed algorithms nor semantic segmentation algorithms with extremely high annotation costs can solve the problem of over/under-segmentation.

2.2. The regressing approach for density map and its application

Similar to the flotation froth image analysis, the images in the crowd density estimation task have occlusions and shadows, and the objects show a dense distribution and high similarity. To solve the above problems, several scholars have presented density map regression-based methods for counting people and estimating density distribution [34,35]. This method has also attracted increasing attention in other fields, such as agriculture [36], medicine [37], and urban traffic [38]. To the best of our knowledge, most existing studies regard a density map as a relay, concentrating on the accuracy of counting rather than the quality of the density map. Only Sindagi and Patel [39] optimized the performance of the crowd counting model from the perspective of improving the quality of the density map. When applying the density map regression method to flotation image analysis tasks, a high-quality density map means more accurate information about the spatial distribution of bubbles, which is helpful for the subsequent extraction of morphological features.

In contrast to the classification-based baseline, the density map regression-based method not only predicts the number of bubbles but also obtains information on the bubble density distribution. In addition, this method does not cause the problem of over/under-segmentation. Thus, the prediction results are more stable.

2.3. The contribution of this study

In this work, we aim to introduce a novel approach for robustly analyzing flotation froth images, and the main contributions are summarized as follows:

The density map regression was first introduced to analyze flotation froth morphological characteristics, and the regression framework was designed to fit flotation froth images.

The ground truth was processed by label normalization, the network was improved by employing the deconvolution module and loss drop operation, and the effectiveness of the above operations was verified via the ablation experiments.

Our method was compared with other segmentation baselines, and its generalization ability was demonstrated on the coal and other minerals flotation froth images.

3. Methodology

3.1. Overview

The framework of density map regression proposed in this paper is shown in Fig. 2. Bubbles were annotated with points. The density map was generated as ground truth based on the position relationship between the points. The images (image-input) and density maps (ground truth-input) were fed into the network for training to obtain the predicted density map. Subsequently, all density values of prediction were summed to get the number of bubbles in one image. The ratio of the area between the large and small bubbles in the image, as well as their spatial distribution, was determined based on the density map's distribution.

3.2. Density map generation

In the dataset production phase, we annotated points on the bubbles of the flotation froth image and generated a two-dimensional matrix of the same size as the image resolution. The value at the point annotated in a matrix is 1, and the value at the rest of the positions is 0. If there is a bubble at the pixel x_i , the matrix with N bubbles annotated would be represented as:

$$D(x) = \sum_{i=1}^N \delta(x - x_i) \quad (1)$$

and $\delta(x - x_i)$ can be represented as:

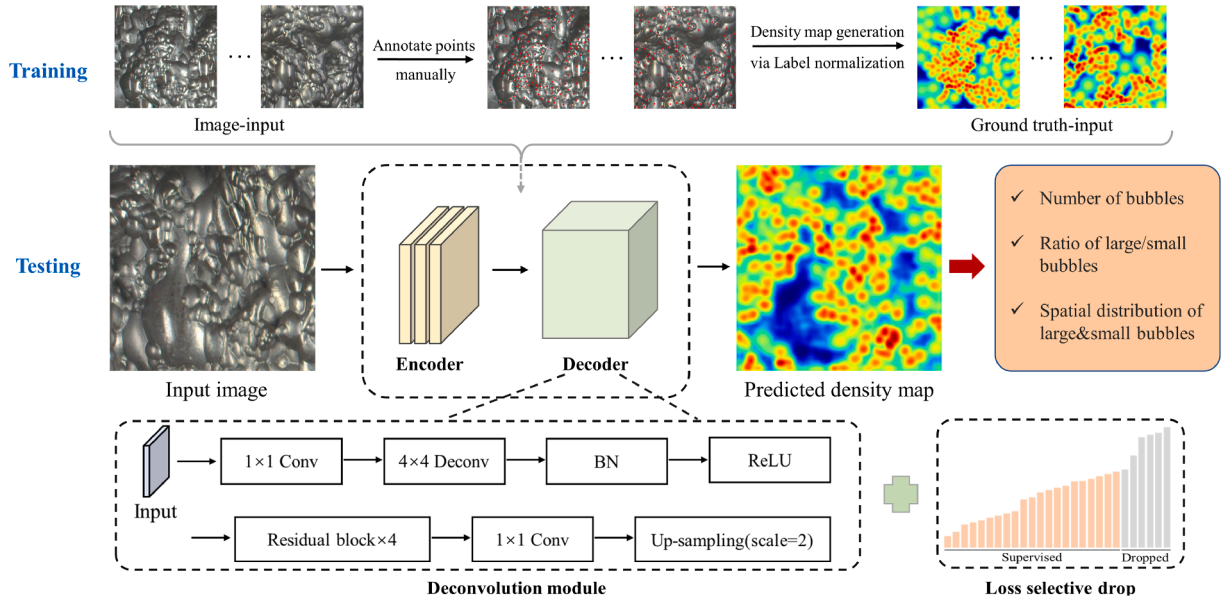


Fig. 2. Overview of density map regression framework.

$$\delta(x - x_i) = 0, x \neq x_i \quad (2)$$

$$\sum_a^b \delta(x - x_i) = 1, a < x_i < b \quad (3)$$

where $[a, b]$ represents the vector space of the annotated binary image. The above two-dimensional matrix is the density map described in this paper.

If Eq. (1) was employed to generate a density map, a very sparse matrix would be obtained, as shown in Fig. 3(a) (i.e., there is only one non-zero element in a 5×5 matrix). Generally, the zero elements were defined as negative samples, and the non-zero elements were defined as positive samples. Regarding a sparse matrix like Fig. 3(a) as the constraint target for regression might introduce a severe positive and negative sample imbalance problem into the training, which was detrimental to the convergence of the network. Therefore, the Gaussian kernel G_σ was used to convolve with Eq. (1) to smooth out the density map, the new density function was $F(x) = D(x) * G_\sigma$. A matrix like Fig. 3(b), taken as constrained target during regression, could alleviate the imbalance between positive and negative samples.

The computing above density equation $F(x)$ assumed that all of the bubbles were the same size, but in reality, the size of the bubbles varied greatly. The Gaussian standard deviation should be set following each bubble's size to preserve the size information of the bubbles on the generated density map. To address this issue, Zhang et al. [40] proposed a density map generation method using a geometrically adaptive Gaussian kernel, where the Gaussian standard deviation was determined by the average distance of the object from K nearest objects. If x_i is a bubble in the image, the K nearest bubbles to it and the average distance can be denoted as $\{d_1^i, d_2^i, \dots, d_K^i\}$ and $\bar{d}_i = \frac{1}{K} \sum_{j=1}^K d_j^i$ respectively. The result of density map generation based on adaptive Gaussian kernel is illustrated in Fig. 4. Thus, the pixel associated with x_i is a region of the scene with a radius proportional to $\bar{d}_i$, and the Gaussian standard deviation should be represented as $\sigma_i = \alpha \bar{d}_i$ and $\alpha = 0.4$. The density equation should be:

$$F(X) = \sum_{i=1}^N \delta(x - x_i) * G_{\sigma_i}(x), \text{ with } \sigma_i = \alpha \bar{d}_i, \quad (4)$$

When we directly utilized the density map generated by Eq. (4) as the ground truth, the regression network found was challenging to maintain a stable convergence (more information was presented in the [supplementary materials](#)). We attempted to multiply the density map by a magnification parameter β to address this issue. We found that the network's training curve was smoother than before, and the regression error was lower. Magnification parameter β is a hyperparameter. We referred to the density map processing method as Label Normalization (LN). Its workflow was as follows:

The original density map was obtained using the generation method described in Eq. (4).

The original density map was multiplied by the magnification parameter and then used as the ground truth to feed into the network for training.

The network output was divided by the magnification parameter to obtain the predicted density map.

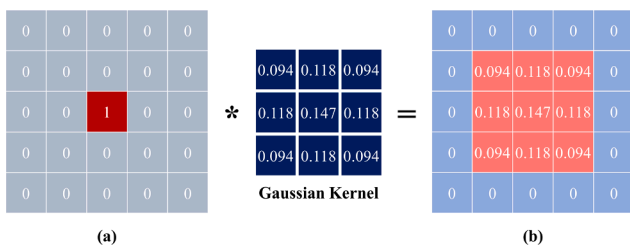


Fig. 3. The process of the convolution of a single annotated bubble using the Gaussian kernel.



Fig. 4. The process of density map generation by using adaptive Gaussian Kernel. In the density map, the warmer color means higher density.

3.3. Network architecture

The network was composed of an encoder and a decoder, and any feature extraction network with simple modification theoretically was qualified as the encoder. The decoder was used to process the semantic information output from the encoder and outputted density maps of the original image size. In addition, loss selective drop mechanism was introduced to reduce the negative impact of annotation deviation when calculating loss.

a. Encoder

In order to make the details of the density map regression as accurate as possible, the encoder needs to achieve high-resolution feature output. HRNet [41] can maintain multiple-resolution feature maps in parallel throughout the network and fuse semantic information at different resolutions. Hence, this study adopted HRNet as the encoder. As shown in Fig. 5, HRNet started from a stem decreasing the resolution to 1/4 of the input image. From the first to the third stage, a new branch was forked from current branches in parallel with 1/2 of the lowest resolution. After four stages of feature extraction, three branches were enlarged through the bilinear up-sampling to the resolution of the high-resolution branch. Finally, we concatenated the representations from all branches and used them as the output of the encoder.

b. Decoder

To extract more details of the bubble boundaries and better characterize the spatial distribution of bubbles, it is advisable to enlarge the resolution of the network output. Although utilizing merely bilinear up-sampling is a straightforward strategy, it is too rough for the regression task, which would cause tearing at the bubble boundaries in the predicted density map and bias the spatial density estimates of the bubbles. Thus, we designed a deconvolution module as the decoder to make the predicted density map smoother.

In this paper, we set the resolution of the network output to the size of the input image. The implementation of a deconvolution module is shown in Fig. 6. A deconvolution module's input was the output feature map of the encoder part, and its output was a single channel image (density map) with the input image size. A deconvolution module consisted of 1×1 convolution, deconvolution part, residual blocks, and a bilinear up-sampling operation. Specifically, we used a 4×4 deconvolution followed by Batch Normalization and ReLU to enlarge the resolution of the feature map from 1/4 to 1/2 (size of the input image), four residual blocks were stacked to refine the enlarged feature maps, 1×1 convolution was added to adjust the number of channels, and a bilinear up-sampling was used to enlarge the resolution to the size of the input image.

c. Loss selective drop

As shown in Fig. 7, there would be a slight deviation in point annotation due to annotator fatigue (subjective reason) and low image quality (objective reason). Still, its annotation cost and the possibility of deviation are much lower than that of segmentation annotation.

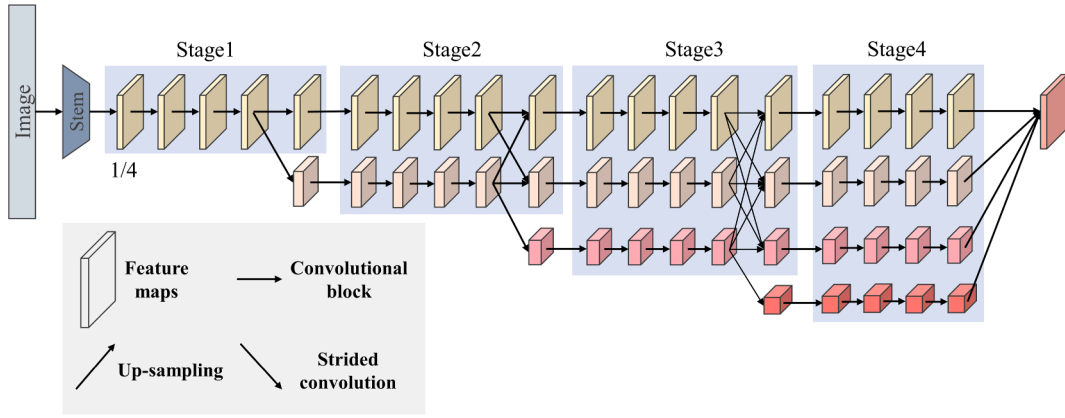


Fig. 5. The architecture of HRNet.

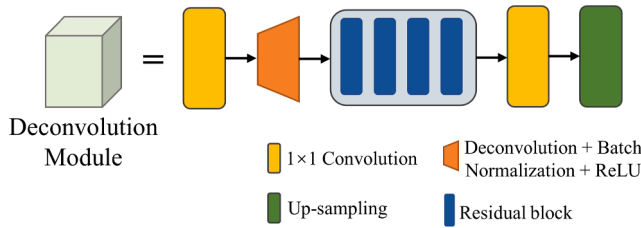
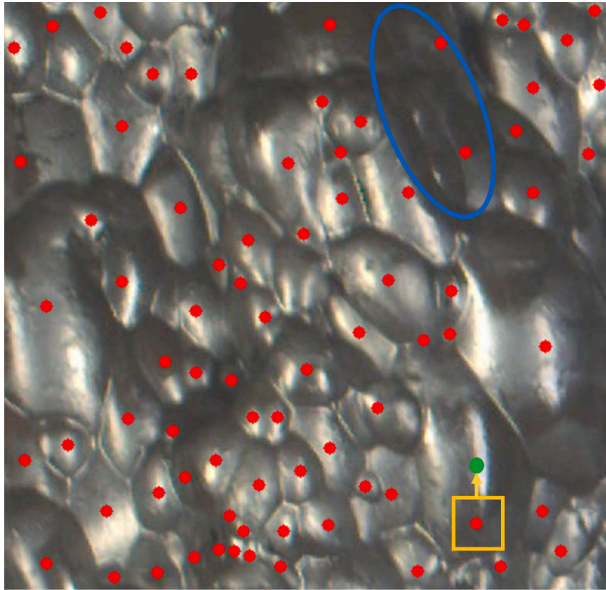


Fig. 6. The structure of the deconvolution module.



— Shadow and black fields ● Correct point labeling position
 — Point labeling deviation

Fig. 7. Flotation froth image with point annotation.

To mitigate the negative impact of this deviation on the model, we optimized the treatment of crowd density estimation [42] and introduced a deviation-selection mechanism named Loss Selective Drop (LSD) in this paper. The definition is as follows:

$$\mathcal{L}_{LSD} = \sum_j^N m^j \bullet \epsilon^j, \quad (5)$$

where N equals the size of the density map, m^j is the binary vector to

measure whether to remove j_{th} deviation, and ϵ^j is j_{th} the deviation between predictions and ground truth, SmoothL1 [43] is the deviation function defined as:

$$\epsilon^j = \begin{cases} 0.5(D_j^{pred} - D_j^{gt})^2, & \text{if } |D_j^{pred} - D_j^{gt}| < 1 \\ |D_j^{pred} - D_j^{gt}| - 0.5, & \text{otherwise} \end{cases}, \quad (6)$$

where D_j^{pred} and D_j^{gt} represent the predictions and ground truth for given the position j , respectively.

If ϵ^j is too large, which means too many loss factors will be ignored, leading to additional bias. To reduce the negative impact of this situation on the further convergence of the model, LSD provides a mechanism to remove large deviations dynamically. We acquired the indices that sort the deviations in ascending order $\vec{k} = \text{sortID}(\epsilon)$. The definition of m^j is as follows:

$$m^j = \begin{cases} 1, & \text{if } j \in \{\vec{k}(1), \vec{k}(2), \dots, \vec{k}(\lfloor \rho N \rfloor)\} \\ 0, & \text{otherwise} \end{cases}, \quad (7)$$

where ρ is the threshold, and we set $\rho = 0.9$, which means only 90 % density value with the lowest deviations will be involved in supervision.

4. Experiments

4.1. Minerals

Flotation froth images were collected from a coal preparation plant in Shandong Province, China. The 1/3 coking coal is used for flotation. In the coal slurry recovery section of the plant, the hydrocyclone is used to concentrate and classify the magnetic separation tailings. The coal slurry from the overflow is pre-treated and fed to the jet flotation machine. The image acquisition device is installed above the second cell of the jet flotation machine.

The flotation froth image acquisition system consists of four parts: an auxiliary illumination system, fixed focus lens, industrial camera, and mechanical support structure (Fig. 8). To ensure light intensity for image acquisition and minimize reflections on the surface of the flotation froth, two auxiliary lighting units are placed at an angle of 30° to the industrial camera. An industrial camera (acA1440-220uc, Basler) is employed for image acquisition, which reached a shooting rate of 227 fps with an array size of 1440 × 1080. The lens used for image acquisition has a focal length of 8.5 mm and an aperture range of F2.4 - F16. The distance between the industrial camera and the flotation froth is 550 mm, and the actual field of view of the captured image is 339 mm × 251 mm.



Fig. 8. Flotation froth image acquisition system.

4.2. Dataset

The coal flotation froth datasets were obtained under normal production conditions at the coal preparation plant. After cropping and data augmentation of the acquired images, the total dataset consisted of 1020 flotation froth images with a size of 512×512 . The bubbles in each image were annotated by point. To mitigate the annotated mistakes from fatigue, we employed different individuals to annotate different parts of the same image. Finally, we divided the dataset into a training set, a validation set, and a test set in a ratio of 8:1:1 (The production of the dataset and its descriptive statistical information was presented in the [supplementary materials](#)).

4.3. Metrics

In this section, we presented evaluation metrics for the regression networks, including count and quality metrics for density maps regressed by the network. The quality is measured using Structural Similarity (SSIM):

$$SSIM(pred, gt) = \frac{(2u_{pred}u_{gt} + C1)(2\sigma_{predgt} + C2)}{(u_{pred}^2 + u_{gt}^2 + C1)(\sigma_{pred}^2 + \sigma_{gt}^2 + C2)} \quad (8)$$

where u_{pred} and u_{gt} are the mean values of the predicted density map and ground truth, respectively; σ_{pred} and σ_{gt} are the variances predicted density map and ground truth, respectively, and σ_{predgt} is their covariance; $C1$ and $C2$ have dynamic values, and they are used to maintain the stability of the equation about calculating SSIM. In this study, these values were calculated by the following equation:

$$C1 = (k_1L)^2, C2 = (k_2L)^2, L = \max(gt) - \min(gt) \quad (9)$$

where $\max(gt)$ represents the max value of ground truth, and $\min(gt)$ represents the min value. The default value of k_1 is 0.01 and of k_2 is 0.03. The closer the SSIM is to one, the more similarity between the predicted density map and ground truth.

The count error is measured using Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{C_i^{pred} - C_i^{gt}}{C_i^{gt}} \right| \quad (10)$$

where N , C_i^{pred} and C_i^{gt} represent the number of the test image, the number of predicted bubbles, and the number of bubbles in the ground truth, respectively. MAPE is a relative error measurement, representing the percentage of predicted number error in the ground truth.

In measuring the accuracy of the morphological characteristics of the bubbles, it is essential to evaluate the bubbles' size objectively. Therefore, we conceived metrics to evaluate the accuracy of mean bubble size. Before performing the calculation of the mean bubble size, it was necessary to convert the units of bubble size from "pixels" (in the image) to " cm^2 " (in the actual area). We first divided the actual area of the captured flotation froth image by the total number of pixels in the image to get the size scale and then converted the bubble size to actual area units. In this paper, the mean size function is $S = S_t/C$, where S_t is the actual area of the flotation froth image and C is the number of bubbles based on the summation of all values in the density map. The accuracy of mean bubble size is measured by E_m :

$$E_m = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{S_i^{pred} - S_i^{gt}}{S_i^{gt}} \right| \quad (11)$$

In the above formula, N is the number of the test image; S_i^{gt} is the actual mean bubble size that is the actual area of the image divided by the number of points annotated; S_i^{pred} is the measured mean bubble size calculated by the predicted density maps. The closer the E_m to zero, the more accurate the mean size of the bubble.

In addition, to quantify the ratio of the area between large and small bubbles as well as the spatial distribution characteristics, it is necessary to define the size of large and small bubbles. In this paper, we defined bubbles larger than 1100 pixels (2.7 cm^2) as large and the rest as small. It is worth noting that " 2.7 cm^2 " is not a precise definition of large and small bubble sizes but a cut-off to facilitate our study of large and small bubble size characteristics. The accuracy of the ratio of the area between large and small bubbles is measured by E_p :

$$E_p = \frac{1}{N} \sum_{i=1}^N |P_i^{pred} - P_i^{gt}| \quad (12)$$

where N is the number of the test image, P_i^{gt} is the actual percentage of large bubble area to the total area determined by the manually annotated segmentation boundary, and P_i^{pred} is the predicted percentage of large bubble area to total area.

Finally, we used PA (Pixel Accuracy) and mIoU (Mean Intersection over Union) as evaluation metrics for the spatial distribution of large and small bubbles.

4.4. Implementation details

All models were implemented with Pytorch 1.8.0 and were trained on NVIDIA 3090 GPU. In addition, the programming language and its version were Python 3.7.11.

This paper used the SGD optimizer with a momentum of 0.95 and a learning rate of 0.0001. The cyclical learning scheduling strategy [44] was used to reduce the learning rate. Finally, the LSD mechanism was adopted to filter deviation, and the total training epoch is 300.

5. Results and discussion

5.1. Ablation studies of network

Several ablation studies were conducted to analyze the effectiveness of different optimization modules, as explained in Sections 3.2 and 3.3.

a. Impact of label normalization.

To demonstrate the role of LN in network optimization, we compared density maps regressed by six different networks. In the case of using LN, considering that magnification parameter β would affect the quality of the density maps and the accuracy of predicting the number of bubbles, we chose four different β to train six various networks, respectively.

As shown in Fig. 9, density maps regressed by the networks without LN have low similarity to ground truth, and valuable information on bubble sparsity cannot be obtained. Hence, extracting morphological

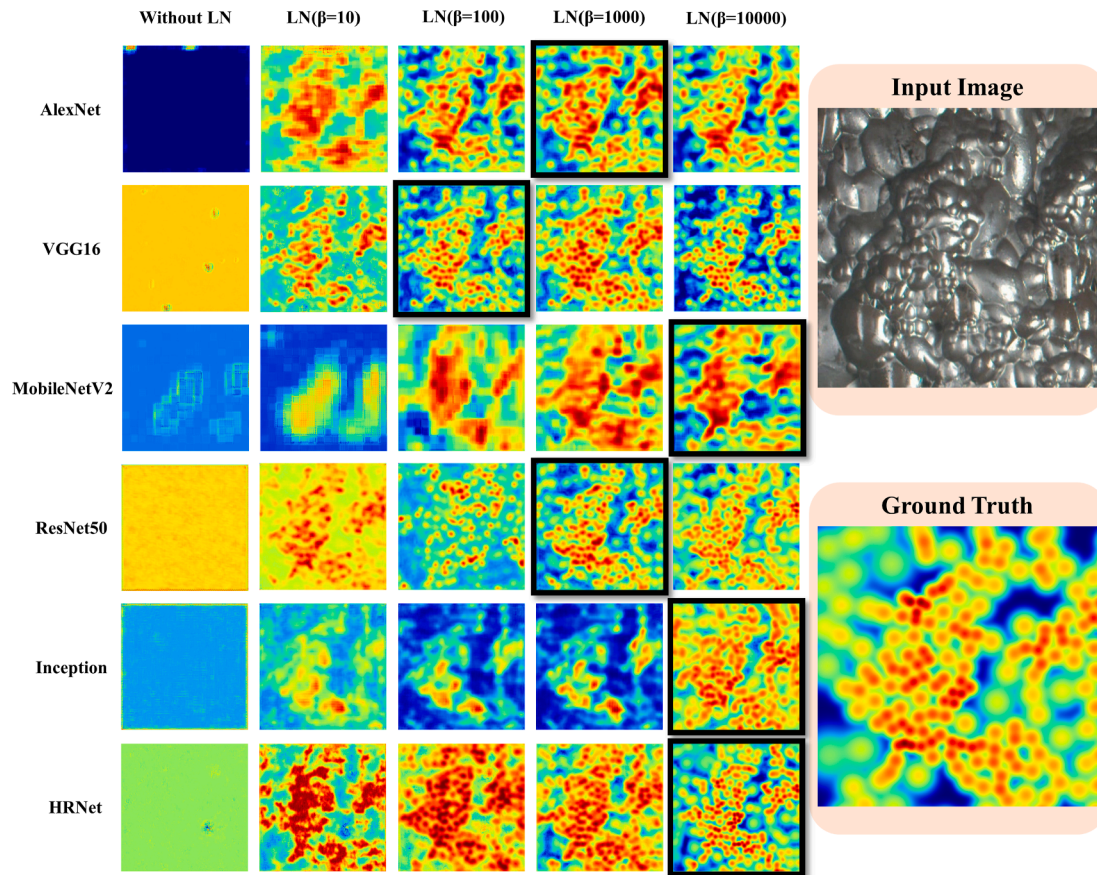


Fig. 9. Comparison of density maps from six different networks with and without LN, and the black bounding box denoted the best result.

bubble characteristics from the density map was difficult. In contrast, the performance of the network trained using LN was better. When β was equal to 10, the network can provide rough bubble density results, but it was not sufficient as a basis for bubble spatial distribution feature extraction. This problem manifested in the form of medium-density bubble regions around small bubbles being mistaken for high-density areas, and almost all low-density bubble regions not being accurately identified. As the value of β increased, the quality of the density maps was considerably improved. When β was equal to 1000 or 10,000, clear density boundaries can be identified in the results of network regression, and the spatial density distribution of bubbles was close to the actual distribution. However, this performance boost plateaued once its β reached a specific value. In this paper, the optimal amplification parameter is the value of β , which predicts the number of bubbles with the highest accuracy.

Based on high-quality density map regression results, Accurate information on the ratio of the area between large and small bubbles can be obtained, which can better identify the health status of the reagent addition. In addition, we can extract the bubble density distribution characteristics to monitor the accumulation of small bubbles.

The value of β was specified as the optimal amplification parameter in the ablation experiments in this section, as well as the discussion of the deconvolution module validity in subsection 5.1b.

Six different feature extraction networks were trained with and without LN to prove the improvement of LN for density map regression. Table 1 demonstrates that the networks using LN reduce bubble count error (MAPE) while achieving better quality in terms of SSIM. The increased accuracy of bubble counting meant that the mean size of bubbles was closer to the actual value. We can indirectly monitor flotation conditions, especially the addition of reagents, by measuring the mean size of bubbles. We can use the bubbles' mean size combined

Table 1

Comparison of different networks with LN and without LN.

Network	Label Normalization	SSIM	MAPE
AlexNet	/	0.52	30.4
VGG16		0.52	30.5
MobileNet-V2		0.52	31.3
ResNet50		0.53	34.8
Inception		0.52	32.4
HRNet	√	0.52	30.2
AlexNet		0.64	15.7
VGG16		0.67	14.3
MobileNet-V2		0.59	18.3
ResNet50		0.67	12.4
Inception		0.62	16.8
HRNet		0.66	11.1

with the bubbles' spatial density distribution and proportional characteristics to describe the morphology of the flotation froth surface appearance in more detail. This laid the foundation for future investigations into the relationship between the morphological characteristics of flotation froth and the flotation process conditions.

b. Impact of deconvolution module.

We performed an ablation study to demonstrate the effects of the deconvolution module. Table 2 illustrates the deconvolution module has effectively improved the density map regression ability of the network. After adopting the deconvolution module as the decoder, six different networks not only improved the quality of the density map but also reduced the error of bubble number prediction.

To better describe the bubble distribution within the images, we matched the range of large and small bubble density values based on the definition of large and small bubble sizes in Metrics. Then we visualized the spatial distribution of large and small bubbles based on the predicted

Table 2

Comparison of different networks using bilinear up-sampling and deconvolution module.

Network	Method	SSIM	MAPE
AlexNet	Bilinear up-sampling	0.58	21.7
VGG16		0.6	20.3
MobileNet-V2		0.55	29.6
ResNet50		0.64	18.9
Inception		0.61	20.9
HRNet	Deconvolution Module	0.65	15.7
AlexNet		0.64	15.7
VGG16		0.67	14.3
MobileNet-V2		0.59	18.3
ResNet50		0.67	12.4
Inception		0.62	16.8
HRNet		0.66	11.1

density map.

As demonstrated in Fig. 10, under the premise of using the same feature extraction network (ResNet50), compared with the bilinear up-sampling, the shape of bubbles was more precise, and the boundary was smoother in the density map of deconvolution module regression. Two quintessential examples (the blue and green coils in Fig. 10) were illustrated to show our optimization results explicitly. The tearing of bubble boundaries in the density map caused by bilinear up-sampling can seriously affect the shape and area of the small bubble region (the red area in Fig. 10(a) & (b)). After replacing bilinear up-sampling with a deconvolution module, which benefited from the optimization of bubble boundary details, the regression area of small bubbles matched better with the area of actual small bubbles in both shape and area (the red area in Fig. 10(c) & (d)).

The above phenomenon proved that the deconvolution module had guided the network to gain higher-quality density maps, which enabled the network to regress the spatial distribution of large and small bubbles accurately. In addition, HRNet with the deconvolution module achieved the best accuracy in identifying the number of bubbles. Hence it was chosen as the baseline of subsequent optimization.

c. Impact of loss selective drop.

We conducted ablation experiments to verify the effectiveness of loss selective drop and the choice of threshold. We compare the counting accuracy and quality of density maps under different thresholds, and the result is shown in Fig. 11. We set the network without loss selective drop as the baseline (the blue dashed line in Fig. 11), which also means $\delta = 1$.

When we set $0.85 \leq \delta < 1$, the accuracy of counting bubbles was much better than supervision by all annotations, and reached the optimum at $\delta = 0.95$; the quality of density maps fluctuated above and

below the baseline, it also peaked at δ equal to 0.95. We observed that the counting accuracy and quality of density maps reached the optimum when $\delta = 0.95$, the SSIM reached a maximum of 67 % and the MAPE achieved an error of 9 %. As for $\delta < 0.85$, the network's performance declined, and it can be attributed to the inadequate usage of ground truth. It might indicate that about 15 % of annotations with significant differences from the actual labels negatively affect the network's performance when involved in training. As demonstrated by ablation experiments, the loss selective drop mechanism reduced this negative influence conveniently and effectively.

5.2. Comprehensive evaluation of flotation froth characteristics extraction

To better verify the effectiveness of the proposed method in this paper, we conducted a comprehensive evaluation of the best model with the following setup, HRNet with deconvolution module, $\beta = 10000$ in LN, and $\delta = 0.95$ in loss selective drop.

a. Comparison with existing methods

The method proposed in this paper was based on the regressed density map to predict the number of bubbles, the ratio of the area between large and small bubbles, and the spatial distribution of large to small bubbles in flotation froth images. Compared to classification methods that focus on the identification of flotation conditions, the method in this paper can obtain more detailed flotation froth characteristics. In addition, considering that segmentation methods can get similar results, we chose two main bubble segmentation methods (watershed algorithm and Swin Transformer [45]) for comparison experiments with the method proposed in this paper.

As the results are shown in Table 3, two extracted significant features (the mean size of bubbles & the ratio of the area between large and small bubbles) were competitive with the other two methods. The density map was trained as a constraint, and our network emphasized counting and localization information of bubbles to understand the froth images better. Therefore, it was more advantageous to use our method regarding tasks such as calculating the mean size and the ratio of the area between large and small bubbles. Furthermore, although our approach was not better than the segmentation method in terms of the spatial distribution index of bubbles, our approach was less expensive regarding annotation.

As shown in Fig. 12, the counting results of the bubbles in the four figures show that our method was more advantageous than that obtained by the segmentation method. Taking Fig. (a) and (b) as examples, under-segmentation in the segmentation method led to easily counting

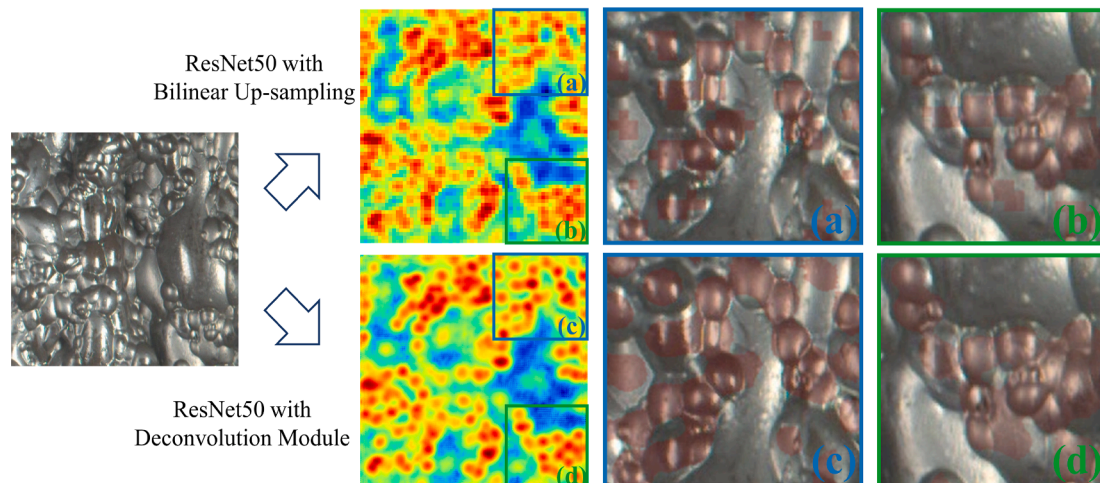


Fig. 10. Example results by ResNet50 with different up-sampling.

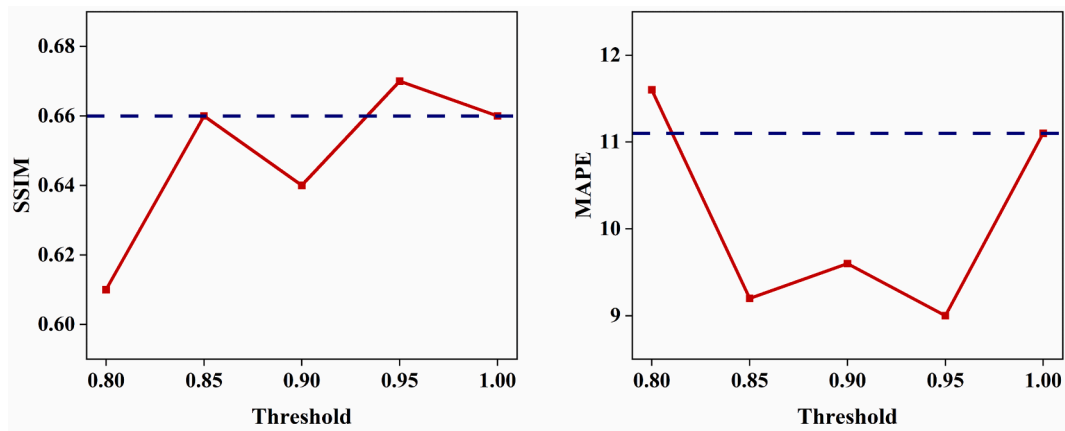


Fig. 11. Effect of threshold δ , the dotted line represents our network without loss selective drop.

Table 3

Comprehensive quantitative comparison with different methods.

Method	Error in measuring bubble mean size	Error in measuring the ratio of the area	Accuracy of measuring the spatial distribution of large and small bubbles			
	E_m (%)	E_p (%)	PA	IoU _{small}	IoU _{big}	mIoU
Watershed algorithm	11.4	16.01	0.74	0.74	0.26	0.5
Swin Transformer	17.1	5.31	0.79	0.77	0.32	0.54
Ours	1.7	0.8	0.77	0.74	0.31	0.53

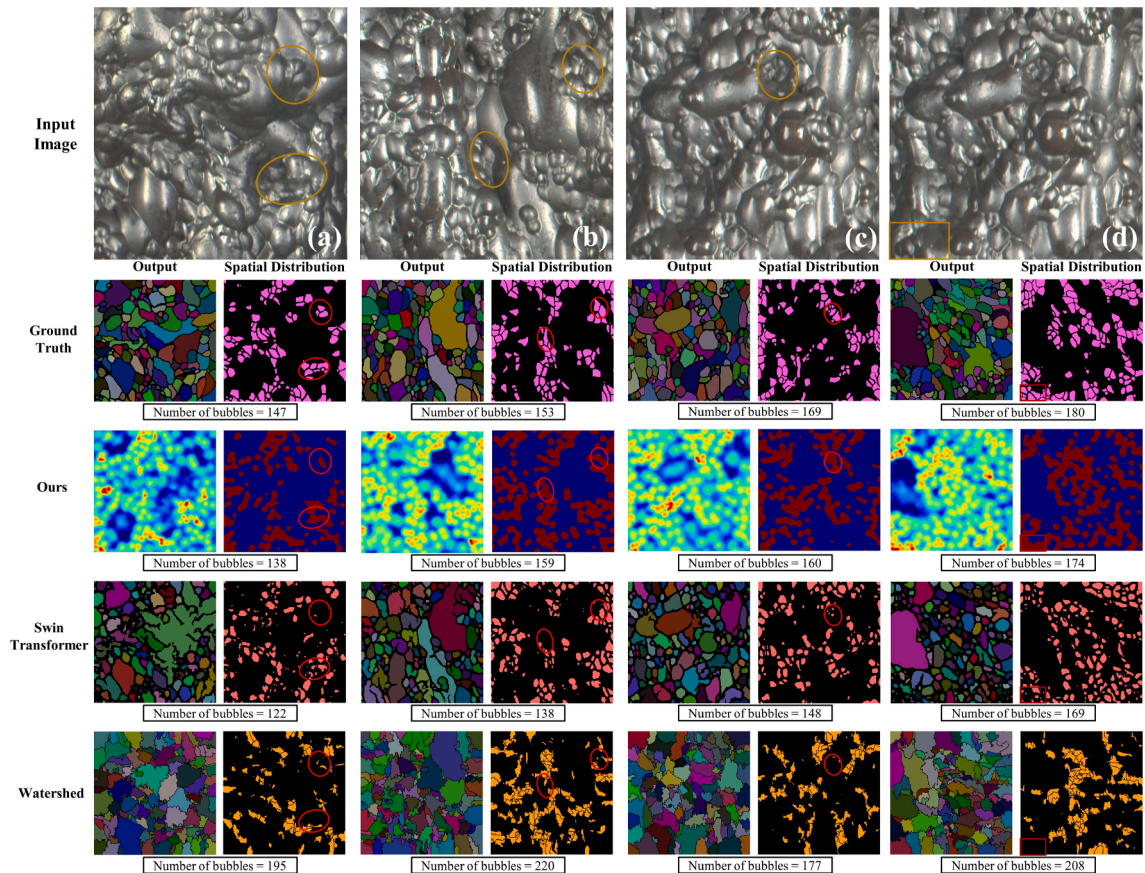


Fig. 12. Comparison of results from different methods in the presence of shadow occlusion. The yellow coils represent the shaded area in the flotation images. Divided by the definition of large and small bubbles in the Metrics, the image in the column of spatial distribution, the colored areas represent the distribution of small bubbles, and black and blue represent the areas of large bubbles.

multiple bubbles into one, which was responsible for the significant decrease in the counting results. From the bubble position distribution, the over-segmentation problem also led to the annexation of multiple bubbles. The binarization process led to the loss of position information of small bubbles (the region in red circles in Fig. 12). In addition, from Fig. 12(d), it was difficult for segmentation-based methods to extract the position information of tiny bubbles, especially those appearing in the shadow or blurred areas. However, it had to be admitted that our method cannot extract the edge information of bubbles due to little annotation information. The problem was easily seen in Fig. (c), where our method had difficulty fitting the shape of the bubbles. Thus, the resulting bubbles were almost circular. Therefore, the indicator of bubble position distribution in Table 3 cannot enjoy the effective gain from accurate counting.

b. Generalization capability of other mineral flotation froth

We collected flotation froth images of lead-zinc [46], nickel ore [47], and coal [48] from the internet to test the domain adaption capability of our model (trained with the dataset in subsection 4.1). The results are shown in Fig. 13.

Without annotation and retraining, our network generated high-quality density maps with the precise density distribution of large and small bubbles. Due to the lack of annotation information, we did not quantify the results for other mineral flotation bubbles. Still, the visualization results can demonstrate the effectiveness of our method in identifying morphological bubble characteristics.

The above phenomenon shows that our method achieved satisfactory results in other mineral flotation scenarios without annotation and retraining. It demonstrates the potential of our method to easily migrate

to similar complex industrial tasks with little or no annotation.

6. Conclusion

In this paper, we attempt to analyze the morphology of coal flotation froth images from the perspective of bubble distribution. A density map regression-based framework was proposed to measure the sparsity of bubbles. The method of density map regression was designed for the counting task, and we extended it to the study of fitting density distributions. In order to improve the quality of regressed density maps and the accuracy of bubble morphological characteristics extraction, HRNet with the deconvolution module was selected to achieve high-resolution semantic rich feature extraction; label normalization and loss selective drop were employed to optimize the generation of datasets and training of the network, respectively. Each module's effectiveness in the framework was demonstrated through a series of ablation experiments on the coal flotation froth dataset.

According to the comparison experiments with existing methods, it was more advantageous to adopt our method regarding tasks such as calculating the mean size and the ratio of the area between large and small bubbles. In predicting the spatial distribution characteristics of bubbles, our method can achieve the same accuracy as the semantic segmentation algorithm with a lower cost in terms of annotation. For the shadowing problem in the industrial flotation dataset, our method was more robust in extracting the morphological characteristics of bubbles. It did not cause significant fluctuations in the prediction due to the degradation of acquisition image quality.

The method in this study provided a new idea for a better understanding and analysis of flotation froth images. Future works may attempt to combine the bubble distribution and morphological

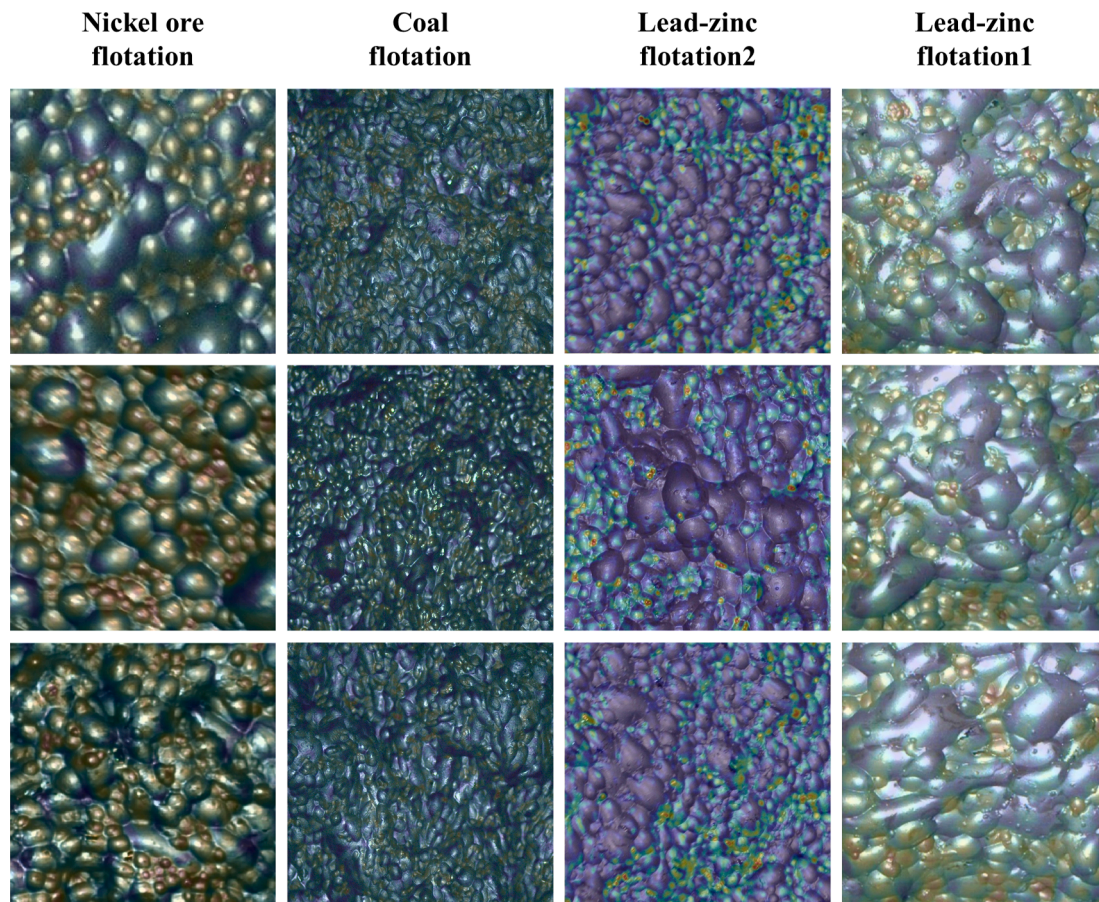


Fig. 13. Example results for flotation froth images of different types of minerals (no annotation and training).

characteristics to propose a more accurate and stable bubble characteristic measurement, which promotes the process of flotation intelligence.

CRedit authorship contribution statement

Yuhan Fan: Methodology, Data curation, Formal analysis, Software, Visualization, Writing – review & editing. **Ziqi Lv:** Methodology, Formal analysis, Writing – original draft. **Weidong Wang:** Conceptualization, Validation, Project administration, Supervision. **Rui Tian:** Formal analysis, Writing – original draft, Writing – review & editing. **Kanghui Zhang:** Data curation, Formal analysis. **Mengchen Wang:** Data curation. **Chenglian Zhang:** Conceptualization, Validation, Project administration, Supervision. **Zhiqiang Xu:** Funding acquisition, Resources, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2022.112212>.

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